U.S. Police Line-of-Duty Deaths

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Abstract:

U.S. law enforcement line-of-duty deaths have been recorded since 1791, but most research on them is descriptive, limited to felonious killings, or based on aggregated counts. This project investigates state-level patterns in line-of-duty deaths by combining historical officer death records from the Officer Down Memorial Page (ODMP) with law enforcement employee-count data from the FBI Crime Data Explorer (CDE). In Semester 1, we will clean and integrate the datasets by state and year, calculate annual death rates per 1,000 sworn officers where reliable officer counts are available, compare trends across states and regions, and evaluate models for forecasting annual death counts by state. Poisson and negative binomial regression, along with appropriate predictive models, will be compared using chronological training and testing periods to prevent data leakage. In Semester 2, we will use natural language processing to analyze incident summaries, identify recurring circumstances surrounding officer deaths, and examine how incident themes and cause-of-death patterns have changed over time, including around major historical events such as September 11, 2001, and the COVID-19 pandemic. This project will produce a cleaned analytical dataset, reproducible code, visualizations, a dashboard, and a final report documenting historical patterns, forecasting performance, and data limitations.

Introduction:

Background Information:

Line-of-duty deaths are among the most serious outcomes in public safety. They harm officers' families, their departments, and the communities they serve. In 2025, 53 officers were feloniously killed in the line of duty, and 81.1% of them died from gunshot wounds [1]. Research shows that lethality depends on situational and encounter characteristics [2], officer age and length of service [3], and community-level conditions such as law enforcement presence and training [4]. Knowing which officer and organizational characteristics are associated with different kinds of fatal outcomes can help agencies set training priorities, plan scheduling, and allocate resources.

Research Gap / Challenges:

Most existing work on this problem is descriptive or regression-based. It typically examines felonious killings only, mostly gunfire, and often uses aggregated counts or short time windows. One national study pooled cross-sectional data to estimate the odds that an assault turns lethal [2]. Another revisited the effects of tenure and age on felonious deaths from 1995 to 2015 [3], and a third analyzed contributing factors at the county level [4]. Supervised machine learning is used in policing, but mostly for other targets. In Charlotte-Mecklenburg, a machine learning model flagged more at-risk officers, with fewer false positives, than the department's existing early-intervention system [5]. That model predicts adverse events involving officer conduct, not officer fatalities.

To our knowledge, few studies combine historical officer death records with state-level officer counts to examine death rates over time while also investigating the circumstances surrounding individual deaths through incident narratives. This project will address this gap by

combining structured death records with employee-count data and examining both historical patterns and predictive model performance.

The data raise several challenges:

- Missing values. Age is missing for 10.4% of records, and the offender field is missing for 46.1%.
- Free-text offender field. It contains values such as “Executed in 1791” and “Acquitted,” so it must be grouped into consistent outcome classes.
- Class imbalance. Gunfire dominates the cause categories, so we will use grouped targets and evaluate classification models with macro-F1.
- Limited predictors. The available variables capture only part of what determines a death, so model performance may be modest.
- Different data coverage. The death records span more than two centuries, while officer-count data cover a more limited period. Death rates can only be calculated for state-year combinations with reliable and comparable officer counts.
- Inconsistent definitions. The two datasets may differ in state identifiers, reporting periods, and agency coverage. These differences must be reviewed before merging.
- Unusual events. COVID-19 and 9/11 produced large numbers of illness-related and multi-officer deaths that could influence historical trends.

Objective(s):

Our main goal is to understand how U.S. law enforcement line-of-duty deaths vary across states and over time, accounting for differences in the size of the sworn officer population where reliable officer counts are available.

Semester 1: Data preparation, classification, and death-rate analysis

1. Prepare and audit the data. Clean and standardize the officer death records and employee-count data. Group the free-text offender field into consistent legal-outcome classes, assign each death to a broader cause type, and flag multi-officer incidents. Standardize state identifiers and reporting years before integrating the datasets.

2. Build and compare classification models. Train decision tree, Random Forest, and XGBoost models against a majority-class baseline to predict (a) cause of death and (b) offender legal outcome. Use a training and testing split with cross-validation on the training data. The primary metric will be macro-F1, and we will report whether each model improves on the baseline on the held-out test set.

3. Analyze officer death rates from 1990 to 2022. Merge the death records with available FBI CDE employee counts and other relevant sources where needed. Aggregate deaths by state and year, calculate death rates per 1,000 sworn officers, and fit Poisson and negative binomial regression models using officer counts as exposure offsets where appropriate. Compare models using dispersion and AIC and report incidence rate ratios (IRRs) with 95% confidence intervals where appropriate.

4. Compare federal and state officer death rates. Use relevant federal officer-count data, including BJS data where available, to compare federal and state death rates per officer, subject to data availability and comparability. Conduct a rate-ratio test at a 0.05 significance level where the assumptions are met.

5. Examine unusual historical events. Rather than automatically dropping anomaly years, calculate and compare death rates by cause type where possible. This will help us examine how the COVID-19 pandemic and delayed 9/11-related illness deaths affected illness-specific and overall trends.

Semester 2: Narrative analysis

We will use natural language processing to analyze incident summaries, identify recurring circumstances surrounding officer deaths, and examine how incident themes have changed over time. We will explore whether narrative information provides context that is not captured by the structured variables and investigate changes around major historical events.

Dataset:

This project will use two datasets to analyze U.S. police line-of-duty deaths and examine how officer population relates to death trends across states and years.

Dataset 1: U.S. Police Line-of-Duty Deaths

The first dataset, police_deaths_USA_v7.csv, contains historical records of U.S. law enforcement officers who died in the line of duty. The data spans from January 3, 1791, through December 4, 2022, and includes information about the officer, department, state, date of death, cause of death, and incident details. The dataset contains 13 columns, including numerical, categorical, date, and text variables. Examples include officer age, rank, state, department, weapon, offender information, and incident summary.

The dataset contains missing values and inconsistencies that must be addressed before analysis. For example, officer age is missing in approximately 10.4% of records, and offender information is missing in approximately 46.1% of records. Other fields, including badge number, weapon, and length of service, also contain missing information. Data cleaning will include checking for duplicate records, standardizing categorical values, converting dates and numerical fields to appropriate formats, and examining unusual or inconsistent entries. The incident summary text will primarily be used for the second semester's text analysis.

Source: Mayuresh Koli, Police Deaths in USA from 1791 to 2022, Kaggle. The records are based on information from the Officer Down Memorial Page (ODMP).



Rank	Name	Age	End_Of_W	Day_Of_W	Cause	Departmer	State	Tour	Badge	Weapon	Offender	Summary
Constable	Darius Quimby		1791-01-0	Monday	Stabbed	Albany Cou	New York			Edged wea	Executed i	Constable Darius Quimby was stabbed to death while attempting to
Sheriff	Cornelius H	53	1791-10-2	Saturday	Gunfire	Columbia (	New York	2 years		Gun; Unkn	Acquitted	Sheriff Cornelius Hogeboom was shot and killed while serving a writ
Deputy She	Isaac Smith		1792-05-1	Thursday	Gunfire	Westchest	New York			Handgun	Executed i	Deputy Sheriff Isaac Smith was shot and killed while attempting to s
Marshal	Robert For	40	1794-01-1	Saturday	Gunfire	United Stat	United Stat	4 years, 4 r	4	Handgun; I	Escaped	Marshal Robert Forsyth, believed to be one of the first U.S. Marshals
Deputy She	Robert Berwick		1797-06-2	Thursday	Gunfire	New York (	New York			Handgun	Executed i	Deputy Sheriff Robert Berwick was shot and killed while attempting
Sheriff	Robert Max	45	1797-11-1	Sunday	Gunfire	Washingto	South Carolina			Gun; Unkn	Acquitted	Sheriff Robert Maxwell was shot and killed after being ambushed wh
Superinter	Hilary Bake	52	1798-09-2	Tuesday	Duty relate	Philadelph	Pennsylvar	1 year, 4 m	SUPERINTEN			Superindentent Hilary Baker died as a result of contracting yellow fe

Dataset 2: FBI Law Enforcement Employee Counts

The second dataset contains 785,127 rows of law enforcement employee-count records. It includes information organized by year and jurisdiction, with fields such as data_year, agency identifiers, agency and county names, state abbreviation, division, region, agency type, population, male and female officer counts, male and female civilian employee counts, total officer count, total civilian count, total personnel count, and personnel per 1,000 population.

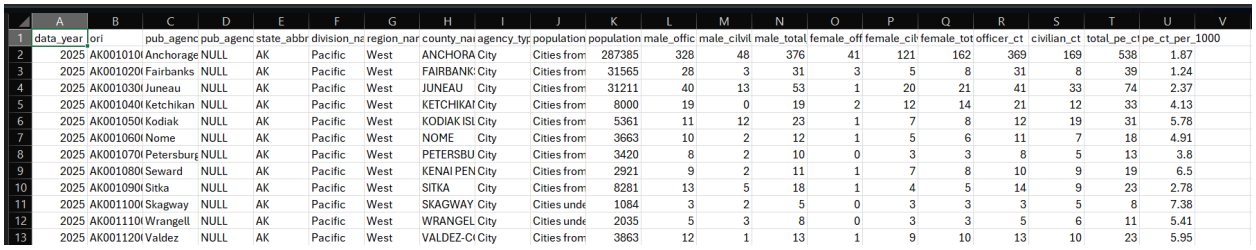
The dataset provides information about the number of law enforcement employees serving in different jurisdictions. For example, the displayed records include data for Alaska agencies in 2025. The dataset contains agency-level records, so the data will need to be aggregated by state and year before it can be combined with the police death dataset. The exact years covered and the number of columns will be confirmed during the data audit.

This dataset is important because the number of officer deaths alone does not account for differences in the size of law enforcement workforces. Combining the employee counts with the death records will allow the project to calculate death rates relative to the number of officers, where compatible data are available. For example, the project may calculate deaths per 1,000 sworn officers for each state and year. The team will verify that the officer-count definition, geographic level, and year match the death records before calculating these rates.

The employee-count dataset also contains fields that require careful handling. Some values may be listed as NULL, and some records may have zero or unavailable population

values. These cases will be checked before calculating rates to avoid division-by-zero errors or misleading results. Because the employee-count data are recorded at the agency level, the team will aggregate compatible records rather than directly merging every agency row with individual death records.

Source: FBI Crime Data Explorer, Law Enforcement Employee Counts.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	data_year	ori	pub_agenc	pub_agenc	state_abbr	division_na	region_nar	county_nai	agency_typ	population	population	male_offic	male_civil	male_total	female_off	female_civ	female_tot	officer_ct	civilian_ct	total_pe_ct	pe_ct_per_1000	
2	2025	AK001010(	Anchorage	NULL	AK	Pacific	West	ANCHORA	City	Cities from	287385	328	48	376	41	121	162	369	169	538	1.87	
3	2025	AK001020(	Fairbanks	NULL	AK	Pacific	West	FAIRBANK	City	Cities from	31565	28	3	31	3	5	8	31	8	39	1.24	
4	2025	AK001030(	Juneau	NULL	AK	Pacific	West	JUNEAU	City	Cities from	31211	40	13	53	1	20	21	41	33	74	2.37	
5	2025	AK001040(	Ketchikan	NULL	AK	Pacific	West	KETCHIKAI	City	Cities from	8000	19	0	19	2	12	14	21	12	33	4.13	
6	2025	AK001050(	Kodiak	NULL	AK	Pacific	West	KODIAK ISL	City	Cities from	5361	11	12	23	1	7	8	12	19	31	5.78	
7	2025	AK001060(	Nome	NULL	AK	Pacific	West	NOME	City	Cities from	3663	10	2	12	1	5	6	11	7	18	4.91	
8	2025	AK001070(	Petersburg	NULL	AK	Pacific	West	PETERSBU	City	Cities from	3420	8	2	10	0	3	3	8	5	13	3.8	
9	2025	AK001080(	Seward	NULL	AK	Pacific	West	KENAI PEN	City	Cities from	2921	9	2	11	1	7	8	10	9	19	6.5	
10	2025	AK001090(	Sitka	NULL	AK	Pacific	West	SITKA	City	Cities from	8281	13	5	18	1	4	5	14	9	23	2.78	
11	2025	AK001100(	Skagway	NULL	AK	Pacific	West	SKAGWAY	City	Cities unde	1084	3	2	5	0	3	3	3	5	8	7.38	
12	2025	AK001110(	Wrangell	NULL	AK	Pacific	West	WRANGEL	City	Cities unde	2035	5	3	8	0	3	3	5	6	11	5.41	
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Combining the Datasets

The two datasets will be cleaned separately before being combined. The death dataset will be summarized by state and year, and the employee-count dataset will be aggregated to the same level. The resulting state-year dataset will be used to compare officer death counts with officer workforce size and calculate rates where sufficient data are available.

The combined dataset will support exploratory data analysis, comparisons across states and years, and count-based predictive modeling. Since the death dataset extends much further back in time than the employee-count data may cover, the analysis will distinguish between historical analyses using the death records alone and rate-based analyses using the years for which compatible employee counts are available. This will help prevent missing workforce data from being interpreted as zero officers or zero deaths.

Methodology:

In this section, we outline the step-by-step approach we will follow for Capstone 1 (predictive modeling and death-rate analysis) and Capstone 2 (narrative analysis). We will model deaths using a state-year-cause-type panel in which each row represents a state, year, and cause-type combination. The number of deaths will be the target for the count models, while officer counts will be used to calculate rates and serve as exposure information where appropriate. Capstone 1 will focus on structured data, while Capstone 2 will incorporate information extracted from the incident summaries.

Data Understanding and Exploration

First, we will perform exploratory data analysis (EDA) to understand both datasets. Since the death records span from the eighteenth century to the twenty-first century, we expect differences in data quality and wording. EDA will help us document those differences. We will count records and columns, examine data types, and count unique values in the death dataset. We will also inspect the employee-count dataset to determine its available years, geographic coverage, employee categories, and missing values.

We will group the recorded causes into broader categories, including violent (such as gunfire, stabbing, and assault), accidental (such as automobile and other vehicular incidents), illness (such as heart attacks, COVID-19, and 9/11-related illness), and Other. We will report the number of deaths in each group.

We will examine officer counts by state and year using the FBI CDE dataset and other relevant sources where needed. We will check whether these sources provide comparable

definitions and sufficient coverage for the planned analysis. The death records and employee counts will be summarized separately before integration to identify missing state-year combinations and possible inconsistencies.

We will use histograms to examine age, line charts to show cause-type trends over time, and bar charts to display the most frequent causes of death by state and rank. We will plot deaths against available officer counts by state and compare trends by cause type across regions. We will also examine whether changes in the size of the officer population correspond to changes in annual death counts.

One of the main challenges is the number of missing values in the death records. We also expect unusual spikes from multi-officer incidents and major historical events such as 9/11 and COVID-19. Officer counts may be unavailable or inconsistent for earlier periods, which could limit the death-rate analysis to more recent years.

Data Preprocessing and Preparation

The following steps will be applied to the death dataset and employee-count dataset before they are combined for the state-year analysis.

Cleaning:

We will standardize column names, check for duplicate rows, and convert ages to numeric values. For unusual ages, we will verify the officer's age using reliable sources where possible and document any corrections rather than automatically deleting records.

For cause-type grouping, we will assign each recorded cause to a broader category. Uncommon causes that do not fit the main groups may be assigned to an Other category. We will document the mapping decisions so they can be reproduced.

For the employee-count dataset, we will standardize state identifiers and reporting years and identify the employee categories needed for the analysis. We will check for duplicate state-year observations and determine how missing or inconsistent counts should be handled.

We will aggregate the death records by state, year, and cause type. We will then merge the aggregated death counts with the employee-count data using compatible state identifiers and years. The merged panel will be checked for unmatched records, missing officer counts, and unexpected changes in row counts.

Where appropriate, officer counts will be used as exposure offsets in count regression models. Annual death rates per 1,000 sworn officers will be calculated only when the numerator and denominator are sufficiently reliable and comparable.

Missing Values:

We will handle missing values according to their usefulness to the project. Columns that are mostly empty or do not carry useful information for a particular model may be excluded from that model. For variables that remain, we will evaluate suitable imputation methods, such as the median for numeric variables and the most frequent category for categorical variables, where appropriate.

Missing officer counts will not automatically be imputed because doing so could produce misleading death rates. Instead, we will investigate the missingness and restrict rate analyses to

observations with appropriate denominators. Missingness will be documented because it may reflect differences in historical recordkeeping and reporting coverage.

Imputation values and other preprocessing steps will be fitted using the training data and then applied to validation and test data to avoid data leakage.

Encoding and Standardizing:

For categorical variables, such as cause type, region, and state, we will use suitable encoding methods, including one-hot encoding where appropriate. Numeric variables, such as year and log-transformed officer counts, may be standardized for models that require scaling.

Tree-based models generally do not require standardization, but preprocessing will be documented for each model. Before modeling, we will also confirm that the employee-count variables represent the intended officer population and are not mixed with civilian employee counts when calculating sworn-officer death rates.

Class Imbalance and Augmentation:

Gunfire and other common causes may be represented much more frequently than less common categories. For classification models, we will evaluate class weights and other appropriate approaches to address imbalance. Macro-F1 will be used as the primary classification metric so that performance on less common classes is considered.

For count models, we will use methods designed for count outcomes rather than treating death counts as ordinary classification labels. We may also examine whether state-year observations with zero deaths differ from those with one or more deaths, if that distinction supports the research question.

Modeling Plan

We will model the state-year-cause-type panel, using the number of deaths as the primary target and available state, year, cause-type, and officer-count information as predictors or exposure information, depending on the model.

Planned Models:

- Baseline: A simple baseline based on historical or average death counts, defined using the training period.
- Poisson regression: A statistical count model for death counts, with officer counts used as an exposure offset where appropriate.
- Negative binomial regression: A count model that can accommodate overdispersion when the variance exceeds the mean.
- Decision Tree: A single tree with limited depth to capture interpretable relationships.
- Random Forest: An ensemble of trees that can capture nonlinear patterns and interactions.
- XGBoost: A gradient-boosting model using a suitable objective for the target being modeled.

Justification

Since each row represents a state, year, and cause type, many rows may have zero deaths. Poisson and negative binomial regression are suitable candidates for modeling count outcomes, with the negative binomial model providing an option when overdispersion is present. Officer

counts can be incorporated as exposure offsets where appropriate so that differences in the size of the officer population are considered in the rate analysis.

Tree-based models can capture interactions between categorical and numerical variables and may identify nonlinear patterns. Random Forest can provide a more stable ensemble than a single decision tree, while XGBoost can improve predictive performance through gradient boosting. The models will be compared empirically rather than assuming that one will perform best.

Implementation Tools

We will use Python with:

- Pandas and NumPy for data handling and preprocessing.
- Scikit-learn for pipelines, decision trees, Random Forest, and evaluation.
- Statsmodels for Poisson and negative binomial regression.
- XGBoost for gradient-boosting models.
- Matplotlib for visualizations and diagnostic plots.
- Google Colab and Jupyter for coding and reproducibility.

Model Training Strategy

Splitting:

We will split the data chronologically rather than randomly so that future years are not used to predict earlier years. The final training, validation, and test periods will be determined

after the data audit and examination of the available years. Older years will generally be used for training, an intermediate period for validation where needed, and more recent years for testing.

Cross-validation will be performed within the training period where appropriate. The final test period will remain separate until model evaluation. We will ensure that any preprocessing, including imputation and scaling, is fitted using only the training data.

For XGBoost, we will test a reasonable range of boosting rounds with early stopping based on validation performance where supported. For Random Forest, we will test a range of tree counts. Hyperparameters will be selected based on training and validation results rather than the held-out test set.

Evaluation Plan

Regression Metrics:

For models predicting death counts, we will report mean absolute error (MAE), root mean squared error (RMSE), and other suitable measures, such as Poisson deviance. R-squared may also be reported where appropriate. We will compare each model against the baseline on the held-out test years.

For the rate analysis, we will examine regression diagnostics and compare Poisson and negative binomial models using dispersion and AIC where appropriate. If rate ratios are reported, we will include confidence intervals and describe the limitations of the underlying counts.

Classification Metrics:

For models predicting cause type and offender legal outcome, we will report macro-F1 as the primary metric, along with accuracy, precision, recall, and confusion matrices where appropriate. We will compare each model with the majority-class baseline to determine whether it provides meaningful improvement.

Error Analysis:

We will examine the largest prediction errors by state, year, and cause type, with attention to multi-officer incidents, unusual historical events, and states with relatively few recorded deaths. We will also examine which categories contribute most to model errors and whether performance differs between common and uncommon categories.

Plan for Capstone 2 (Next Semester)

We will use the Capstone 1 exploratory analysis and model error analysis to refine the research questions and identify patterns that structured variables do not fully explain. We will then use text analysis to investigate circumstances described in incident summaries, such as traffic stops, warrant services, and ambushes.

Because many offender outcomes are missing, we will explore whether narrative summaries can provide additional information about legal outcomes. Any inferred outcomes will be treated as uncertain and evaluated against records where the outcome is known, rather than automatically treated as confirmed facts.

We will also explore comparisons between federal and state officer deaths using the summaries and a chi-squared test at a 0.05 significance level where the data and test assumptions permit. Topic modeling methods, including NMF and LDA, will be used to identify recurring

themes and examine how they change over time. Additional models may be explored if they support the research questions and are feasible with the available data.

Expected Outcomes

Applications:

This project is intended as an evidence base for officer-safety research and policy, not as a tool for predicting which individual officer will be harmed. Predictors such as age, length of service, rank, and day of the week describe only part of what determines a death, so we expect model performance to be limited and will report this clearly.

If certain tenure, rank, or time-of-week patterns are linked to particular causes of death, agencies and training academies could use the results to decide where to focus safety training and refresher courses. Such findings would be hypotheses to test, not conclusions to act on alone. Descriptive findings by state and time period could help departments and policymakers see where and when fatal risk has historically been concentrated and how it has shifted over time.

Combining historical death records with officer-count data will also allow us to distinguish changes in the number of deaths from changes in the size of the officer population. The cleaned datasets, grouped offender legal-outcome classes, and baseline results will give later researchers a documented starting point for this problem. Because the data span more than two centuries and contain missing fields, results will be reported at an aggregate level with clear limitations so they are not interpreted as predictions about specific officers or departments.

Deliverables:

We plan on producing a cleaned analytical dataset and data dictionary. The dataset will include standardized death records, documented cause-type and offender legal-outcome classes, and state-year death counts. The merged analytical dataset will include officer counts and calculated death rates for years where reliable denominators are available. The data dictionary will define each variable and describe cleaning, aggregation, and merging decisions.

Reproducible code repository. A GitHub repository will contain the full pipeline, including data checks, preprocessing, state-year panel construction, exploratory analysis, baseline models, Poisson and negative binomial regression, decision tree, Random Forest, and XGBoost models, along with the time-based split and evaluation procedures. It will also include a README with instructions for reproducing the results.

Model comparison and results. We will report appropriate metrics on the held-out test set for the prediction targets, with diagnostic plots and feature-importance results where supported. We will compare the models against the baseline and discuss their limitations.

Visualization report or dashboard. A Tableau dashboard or report will show trends by cause, time period, state, and officer characteristics, building on the Semester 1 EDA. Death counts and rates will be shown where reliable officer-count denominators are available.

Semester 2 text-analysis component. We will apply topic modeling, including NMF and LDA, to the narrative Summary field and report how incident themes have changed over time.

Final report and presentation. A final written report and presentation will cover the data sources, cleaning and integration process, methods, findings, limitations, and future directions.

Team Plan

Our team has four members: Jayden Howard, Jennifer Tran, Ruth Slaughter, and Devashish Patel. Each member leads one stage of the pipeline and also reviews the work of the stage before and after theirs. Every member is responsible for the accuracy of the full project, so each will be able to explain and reproduce any part of the analysis.

- Jayden Howard, data audit and panel construction. Audit missing values and inconsistent fields in the death dataset, including Age, Tour, Offender, and mixed date formats. Map causes to the violent, accidental, illness, and Other groups, flag multi-officer incidents, and build the state-year-cause-type panel. Assist with standardizing state identifiers and checking the merged dataset.
- Jennifer Tran, exploratory analysis and state context. Produce the EDA on cause distribution, age, weekday, geography, and long-term trends. Collect and check officer-count and population data from FBI, BJS, BLS, and Census sources where appropriate. Examine state-year coverage and develop the Tableau dashboard.
- Ruth Slaughter, baseline and count models. Build the preprocessing pipeline, baseline, Poisson and negative binomial models, and time-based split. Help evaluate how officer counts can be incorporated into count models and rate comparisons.
- Devashish Patel, tree models, evaluation, and model selection. Build and tune the decision tree, Random Forest, and XGBoost models. Evaluate the models against the baseline and produce diagnostic plots and feature-importance results.

Shared work. All four members will take part in the following:

- Agreeing on the final cause-type mapping, state-year panel definition, data-merging decisions, and split years, including changes after instructor feedback.

- Reviewing each other's code and results before they are reported.
- Verifying the accuracy and consistency of the death counts, employee counts, and calculated rates.
- Writing the final report and presentation and collaborating on the Semester 2 narrative-analysis plan.

References:

[1] Federal Bureau of Investigation, “FBI Releases Officers Killed and Assaulted in the Line of Duty, 2025 Special Report and Law Enforcement Employee Counts,” May 11, 2026. FBI.gov.

[2] “Assaults against U.S. law enforcement officers in the line-of-duty: situational context and predictors of lethality,” 2016. PubMed PMID: 27885587. https://pubmed.ncbi.nlm.nih.gov/27885587/

[3] K. Tucker-Gail et al., “Felonious line-of-duty officer deaths (1995–2015): The impact of tenure and age revisited,” 2022. doi: 10.1177/14613557211032608.

[4] J. M. Doucet and L. A. Eargle, “Law Enforcement Murdered in the Line of Duty: A County-Level Analysis of Contributing Factors,” Criminology, Criminal Justice, Law & Society, vol. 24, no. 2, pp. 9–30, 2023.

[5] Complete citation for the Charlotte-Mecklenburg early-intervention system study discussed in the original proposal. Verify the original source details before submission.

[6] M. Koli, “Police Deaths in USA from 1791 to 2022,” Kaggle. Available: https://www.kaggle.com/datasets/mayureshkoli/police-deaths-in-usa-from-1791-to-2022. Accessed September 9, 2026.

[7] Officer Down Memorial Page, official website and historical memorial records. https://www.odmp.org/. Accessed October 5, 2026.

[8] Federal Bureau of Investigation, Crime Data Explorer, Law Enforcement Employee Counts dataset. Add the exact dataset URL and access date used by the team before submission.